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Semiconductor device package and methods of manufacture

Abstract

A method includes forming a redistribution structure on a carrier substrate, coupling a first side of a first interconnect structure to a first side of the redistribution structure using first conductive connectors, where the first interconnect structure includes a core substrate, where the first interconnect structure includes second conductive connectors on a second side of the first interconnect structure opposite the first side of the first interconnect structure, coupling a first semiconductor device to the second side of the first interconnect structure using the second conductive connectors, removing the carrier substrate, and coupling a second semiconductor device to a second side of the redistribution structure using third conductive connectors, where the second side of the redistribution structure is opposite the first side of the redistribution structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/186,726, filed on Feb. 26, 2021, now U.S. Pat. No. 11,756,945, issued Sep. 12, 2023, which application is hereby incorporated herein by reference.

BACKGROUND
(1) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components, hence more functions, to be integrated into a given area. Integrated circuits with high functionality require many input/output pads. Yet, small packages may be desired for applications where miniaturization is important.

(2) Integrated Fan Out (InFO) package technology is becoming increasingly popular, particularly when combined with Wafer Level Packaging (WLP) technology in which integrated circuits are packaged in packages that typically include a redistribution layer (RDL) or post passivation interconnect that is used to fan-out wiring for contact pads of the package, so that electrical contacts can be made on a larger pitch than contact pads of the integrated circuit. Such resulting package structures provide for high functional density with relatively low cost and high performance packages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1-5** illustrate cross-sectional views of intermediate steps of forming a device package, in accordance with some embodiments.

(3) FIG. **6**A illustrates a top view of the layout of package regions on a wafer substrate in accordance with some embodiments.

(4) FIG. **6**B illustrates a top view of the layout of package regions on a panel substrate in accordance with some embodiments.

(5) FIGS. **7**A-**10** illustrate cross-sectional views of intermediate steps of forming a device package, in accordance with some embodiments.

(6) FIGS. **11-12** illustrate cross-sectional views of intermediate steps of forming a device package, in accordance with alternative embodiments.

(7) FIGS. **13-15** illustrate cross-sectional views of intermediate steps of forming a device package, in accordance with alternative embodiments.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) Various embodiments provide methods applied to, but not limited to, the formation of a device package in which component devices (e.g., integrated fan-out (InFO) packages, chip-on-wafer (CoW) packages, or the like) are bonded on opposite sides of a core substrate to form a system on integrated substrate (SoIS). The core substrate may include an organic substrate, and the core

substrate may be part of a pre-fabricated interconnect structure that comprises pre-soldered controlled collapse chip connection (C4) bumps. Advantageous features of one or more embodiments disclosed herein may include the ability to electrically connect and integrate a higher number of component devices within the device package. In addition, because of the compact structure of the device package, the package components are closer to each other which results in better power and signal integrity with reduced transmission losses. This leads to better computing performance and allows the use of the device package in applications that require faster processing of data and performing of complex calculations at high speeds (e.g., High-Performance-Computing (HPC)).

(11) FIGS. **1-5** and **7**A-**10** illustrate cross-sectional views of intermediate steps of forming a device package **2500**, in accordance with some embodiments. FIGS. **6**A-**6**B illustrate top views of the layout of package regions in accordance with some embodiments. FIGS. **11-12** illustrate cross-sectional views of intermediate steps of forming a device package **5000**, in accordance with some embodiments. FIGS. **13-15** illustrate cross-sectional views of intermediate steps of forming a device package **7500**, in accordance with some embodiments.

(12) In FIG. **1**, a first package region **1000** of a carrier substrate **102** is shown. The carrier substrate **102** may comprise silicon-based materials, such as a silicon substrate (e.g., a silicon wafer), a glass material, silicon oxide, or other materials, such as aluminum oxide, the like, or a combination. In some embodiments, the carrier substrate **102** may be a panel structure, which may be, for example, a supporting substrate formed from a suitable dielectric material, such as a glass material or an organic material, and which may have a rectangular shape. The carrier substrate **102** may be planar in order to accommodate the formation of additional features subsequently formed thereon. The carrier substrate **102** may comprise a plurality of first package regions **1000** on which package components are fabricated. The package regions **1000** may subsequently be singulated.

(13) In FIG. **2**, an adhesive layer **103** is formed on the carrier substrate **102** to facilitate a subsequent debonding of the carrier substrate **102**. The adhesive layer **103** may comprise a polymer-based material, which may be removed along with the carrier substrate **102** from the overlying structures that will be formed in subsequent steps. In some embodiments, the adhesive layer **103** may comprise an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a Light-to-Heat-Conversion (LTHC) release coating. In some embodiments, the adhesive layer **103** may comprise a ultra-violet (UV) glue, which loses its adhesive property when exposed to UV light. In some embodiments, the adhesive layer **103** may comprise pressure sensitive adhesives, radiation curable adhesives, epoxies, combinations of these, or the like. The adhesive layer **103** may be placed onto the carrier substrate **102** in a semi-liquid or gel form, which is readily deformable under pressure. According to some embodiments, the top surface of the adhesive layer **103** may be leveled and may have a high degree of co-planarity.

(14) Next, a polymer layer **105** is formed over the adhesive layer **103**. The polymer layer **105** provides protection to subsequently formed structures. In an embodiment, the polymer layer **105** may comprise polybenzoxazole (PBO), a polymide, a polymide derivative, Solder Resistance (SR), Ajinomoto build-up film (ABF), or the like. The polymer layer **105** may be formed using a spin-coating process, or the like, and may have a thickness that is in a range from 2 μm to 15 μm.

(15) Contact pads **104** and contact pads **106** of redistribution structure **200** are then formed over the polymer layer **105**. In accordance with some embodiments, the contact pads **104** and the contact pads **106** may be formed by initially forming a first seed layer (not shown) of one or more thin layers of a conductive material that aids in the formation of a thicker layer during subsequent processing steps. The first seed layer may comprise a layer of titanium formed using processes such as sputtering, evaporation, PECVD, or the like. A photoresist (also not shown) may then be formed and patterned to cover the first seed layer using, e.g., a spin coating technique. Once the photoresist has been formed and patterned, a conductive material may be formed on the first seed layer. The conductive material may be a material such as copper, titanium, tungsten, aluminum, another metal,

the like, or a combination thereof. The conductive material may be formed through a deposition process such as electroplating, electroless plating, or the like. Once the conductive material has been formed, the photoresist may be removed through a suitable removal process such as ashing or chemical stripping. Additionally, after the removal of the photoresist, those portions of the first seed layer that were covered by the photoresist may be removed through, for example, a suitable wet etch process or dry etch process, which may use the conductive material as an etch mask. The remaining portions of the first seed layer and conductive material form the contact pads **104** and the contact pads **106**. In some embodiments, a pitch of the contact pads **104** and the contact pads **106** may be different to allow conductive connectors having different pitches to be formed on the contact pads **104** and the contact pads **106**. For example, a pitch of the contact pads **104** may be smaller than a pitch of the contact pads **106**. Specifically, conductive connectors **404** formed on the contact pads **104** (described subsequently in FIG. **8**) may have a pitch P**4** that is in a range from about 50 μm to about 250 μm, and conductive connectors **406** formed on the contact pads **106** (described subsequently in FIG. **8**) may have a pitch P**5** that is in a range from about 150 μm to about 1000 μm.
(16) Further, in FIG. **2**, the remainder of the redistribution structure **200** is formed over the carrier substrate **102**, in accordance with some embodiments. The redistribution structure **200** may comprise a first redistribution portion **252** and a second redistribution portion **254**. The first redistribution portion **252** may comprise insulating layers **208**A-D (e.g., insulating layer **208**A, insulating layer **208**B, insulating layer **208**C, and insulating layer **208**D), and redistribution layers **209**A-D (e.g., redistribution layer **209**A, redistribution layer **209**B, redistribution layer **209**C, and redistribution layer **209**D). The second redistribution portion **254** may comprise insulating layers **208**E-F (e.g., insulating layer **208**E and insulating layer **208**F), and redistribution layers **209**E-F (e.g., redistribution layer **209**E and redistribution layer **209**F). In some embodiments, the redistribution structure **200** may have a number of insulating layers or redistribution layers that is in a range from 1 to 20 layers.
(17) As an example to form the redistribution structure **200**, the insulating layer **208**A is formed over the contact pads **104**, the contact pads **106**, and the carrier substrate **102**. The insulating layer **208**A may comprise one or more dielectric materials such as prepreg, resin coated copper (RCC), an oxide (e.g., silicon oxide), a nitride (e.g., silicon nitride), a photo image dielectric (PID), a polymer material such a PBO, a photosensitive polymer material, a molding material, a polyimide material, a low-k dielectric material, another dielectric material, the like, or a combination thereof. The insulating layer **208**A may be formed by a process such as lamination, coating, (e.g., spin-coating), CVD, the like, or a combination thereof.
(18) Openings into the insulating layer **208**A may be may be patterned directly using an exposure and development process to expose the underlying contact pads **104** and the contact pads **106** when the insulating layer **208**A is formed of a photosensitive polymer such as PBO, polyimide, BCB, or the like. In other embodiments, openings into the insulating layer **208**A may be formed using a suitable photolithographic mask and etching process in order to expose the underlying contact pads **104** and the contact pads **106**. For example, a photoresist may be formed and patterned over the insulating layer **208**A, and one or more etching processes (e.g., a wet etching process or a dry etching process) are utilized to remove portions of the insulating layer **208**A.
(19) The redistribution layer **209**A may then be formed to provide additional routing. In an embodiment, the redistribution layer **209**A may be formed using materials and processes similar to the contact pads **104** and the contact pads **106**. For example, a second seed layer (not shown) may be formed, a photoresist placed and patterned on top of the second seed layer in a desired pattern for the redistribution layer **209**A, and conductive material (e.g., copper, titanium, or the like) may then be formed in the patterned openings of the photoresist using e.g., a plating process. The photoresist may then be removed and the second seed layer etched, forming redistribution layer **209**A. In this manner, the redistribution layer **209**A may form electrical connections to the contact

pads **104** and the contact pads **106**.

(20) Additional insulating layers **209**B-D and redistribution layers **209**B-D of the first redistribution portion **252** and additional insulating layers **209**E-F and redistribution layers **209**E-F of the second redistribution portion **254** may then be formed over the redistribution layer **209**A and insulating layer **208**A to provide additional routing. The insulating layers **209**B-F and redistribution layers **209**B-F may be formed in alternating layers, and may be formed using processes and materials similar to those used for the insulating layer **208**A and the redistribution layer **209**A, respectively. The steps described above may be repeated to form the redistribution structure **200** having a suitable number and configuration of insulation layers and redistribution layers. The insulating layers **208**A-F may be formed to each have a thickness in a range from 2 μm to 100 μm. The redistribution layers **209**A-F may be formed to each have a thickness in a range from 2 μm to 15 μm. In some embodiments, the redistribution structure **200** is a fan-out structure. In other embodiments, the redistribution structure **200** may be formed in a different process than described herein.

(21) In some embodiments, the insulating layer **208**E and insulating layer **208**F of the second redistribution portion **254** may be formed differently from the underlying insulating layer **208**A, insulating layer **208**B, insulating layer **208**C, and insulating layer **208**D of the first redistribution portion **252**. For example, in an embodiment the insulating layer **208**A, the insulating layer **208**B, the insulating layer **208**C, and the insulating layer **208**D may be formed of a material such as an Ajinomoto build up film or a prepreg material to a larger thickness, and the insulating layer **208**E and the insulating layer **208**F may be formed from a different material and/or a different thickness, such as by being formed of PBO. However, any combination of materials and thicknesses may be utilized. In some embodiments, the thickness of the insulating layers **208**A-D of the first redistribution portion **252** and the thickness of the insulating layers **208**E-F of the second redistribution portion **254** are different, and the thickness of the redistribution layers **209**A-D of the first redistribution portion **252** and the thickness of the redistribution layers **209**E-F of the second redistribution portion **254** are different. As a result, the nominal impedance of the redistribution layers **209**A-D of the first redistribution portion **252** and the nominal impedance of the redistribution layers **209**E-F of the second redistribution portion **254** is different.

(22) In FIG. **3**, an interconnect structure **300** is shown, in accordance with some embodiments. The interconnect structure **300** is subsequently bonded to the redistribution structure **200** (shown in FIG. **4**). The interconnect structure **300** is formed in a separate process, and can be tested separately so that a known good interconnect structure **300** is used. For example, in some embodiments, the interconnect structure **300** may be individually or batch tested, validated, and/or verified prior to bonding the interconnect structure **300** to the redistribution structure **200** (shown subsequently in FIG. **4**). In some embodiments, the interconnect structure **300** may be an interposer or a “semi-finished substrate” which could either have active and passive devices or else may be free from active and passive devices. The interconnect structure **300** may provide stability and rigidity to the subsequently attached redistribution structure **200** (shown in FIG. **4**), helping to reduce warping. In some embodiments, the interconnect structure **300** may comprise a core substrate **302** having conductive layers disposed on opposite surfaces. The core substrate **302** may be formed of organic and/or inorganic materials and may comprise a pre-impregnated composite fiber (prepreg) material, an epoxy, a molding compound, Ajinomoto build-up film (ABF), an epoxy molding compound, fiberglass-reinforced resin materials, printed circuit board (PCB) materials, silica filler, polymer materials, polyimide materials, paper, glass fiber, non-woven glass fabric, glass, ceramic, other laminates, the like, or combinations thereof. In some embodiments, the core substrate **302** may comprises between 2 and 10 complete layers of material. In other embodiments, the core substrate **302** may comprise a double-sided copper-clad laminate (CCL) substrate or the like.

(23) During the formation of the interconnect structure **300**, openings are formed in the core substrate **302** within which through vias **306** are formed (described below). In some embodiments,

the openings may be formed by a laser drilling technique, mechanical drilling, etching, or the like. Once the openings have been formed, conductive material is deposited to form the routing layer **308** on a side of the core substrate **302** and through vias **306** within the openings in the core substrate **302**. In some embodiments, the routing layer **308** and through vias **306** may comprise a conductive material such as copper, aluminum, combinations of these, or the like, and are formed using a deposition process such as photoresist patterning and plating, blanket chemical vapor deposition, atomic layer deposition, physical vapor deposition, combinations of these, or the like. The deposition process lines or fills the openings to form the through vias **306**, as well as forming the routing layer **308**. Once the conductive material has been deposited, the conductive material may be patterned (in embodiments in which a blanket deposition was performed) or else the patterned photoresist may be removed (in embodiments in which a plating process is utilized). However, any suitable deposition and/or patterning process may be utilized.

(24) Once the routing layer **308** has been formed, a similar process may then be performed on the opposite side of the core substrate **302** to form the routing layer **309** (and/or remaining portions of through vias **306**) on the opposite side of the core substrate **302**. In this manner, the conductive material may be used to form the routing layer **308** and the routing layer **309** on opposite sides of the core substrate **302** and through vias **306** extending through the core substrate **302**.

(25) Optionally, in some embodiments in which the deposition of the conductive material does not fully fill the openings, a remainder of the openings may be filled with a dielectric material **307**. The dielectric material **307** may provide structural support and protection for the conductive material formed along the sidewalls of the openings. In some embodiments, the dielectric material **307** may comprise a molding material, epoxy, an epoxy molding compound, a resin, the like, or a combination thereof. The dielectric material **307** may be formed or placed using a molding process, a spin-on process, or the like.

(26) Continuing with the formation of the interconnect structure **300**, dielectric layers and additional routing layers may be formed over the routing layers **308** and **309** to form routing structures **312** and **316**. The routing structures **312** and **316** are formed on opposite sides of the core substrate **302** and may provide additional electrical routing within the interconnect structure **300**. The routing structure **312** is electrically connected to the routing layer **308** and includes dielectric layers **310**A-C and routing layers **311**A-C. The routing structure **316** is electrically connected to the routing layer **309** and includes dielectric layers **314**A-C and routing layers **315**A-C. Each of the routing structures **312** or **316** may have any suitable number of dielectric layers or routing layers, including more or fewer than shown in FIG. **3**. In some embodiments, one or both of routing structures **312** or **316** may be omitted. In some embodiments, the number of layers of routing structure **312** may be different than the number of layers of routing structure **316**.

(27) In some embodiments, the routing structure **312** is formed by forming the dielectric layer **310**A over the routing layer **308** and the core substrate **302**. In some embodiments, the dielectric layer **310**A may comprise a build-up material, ABF, a prepreg material, a laminate material, another material similar to those described above for the core substrate **302**, the like, or combinations thereof. The dielectric layer **310**A may be formed by a lamination process, a coating process, or the like. In some embodiments, the dielectric layer **310**A may have a first thickness T**1** of between about 5 μm and about 50 μm. Openings are formed in the dielectric layer **310**A that expose portions of the routing layer **308** for subsequent electrical connection. In some embodiments, the openings are formed by an etching process, a laser drilling technique, or the like. Other processes, e.g., mechanical drilling or the like, may also be used in other embodiments. In some embodiments, an optional surface preparation process (e.g., a desmear process or the like) may be performed after the openings are formed.

(28) A conductive material is then deposited to form routing layer **311**A on the dielectric layer **310**A and within the openings in the dielectric layer **310**A. In some embodiments, the routing layer **311**A is formed by first forming a seed layer and a patterned mask over the dielectric layer **310**A.

The patterned mask may be, for example, a patterned photoresist layer. Openings in the patterned mask may expose portions of the seed layer on which conductive material will subsequently be formed. The conductive material may then be deposited on the exposed regions of the dielectric layer **310**A and within the openings in the dielectric layer **310**A using, for example, a plating process, an electroless plating process, or the like. In some embodiments, the conductive material is deposited having a thickness of between about 1 μm and about 50 μm. After depositing the conductive material, the patterned mask layer (e.g., the photoresist) may be removed using a wet chemical process or a dry process (e.g., an ashing process). In this manner, an additional routing layer (e.g., routing layer **311**A) is formed over and electrically connected to the routing layer **308**.
(29) Additional dielectric layers **310**B-C and routing layers **311**B-C may then be formed adjacent to the routing layer **311**A and dielectric layer **310**A to provide additional routing along with electrical connection within the routing structure **312**. The dielectric layers **310**B-C and routing layers **311**B-C may be formed using processes and materials similar to those used for the dielectric layer **310**A or the routing layer **311**A. These steps may be repeated to form a routing structure **312** having any suitable number and configuration of dielectric layers and routing layers.
(30) In some embodiments, dielectric layers **314**A-C and routing layers **315**A-C may be formed adjacent to the routing layer **309** to form the routing structure **316**. The routing structure **316** may be formed using a process similar to that of the routing structure **312**, described above. However, any suitable process may be utilized. In some embodiments, bond pads **322** and bond pads **323** are formed over and electrically connected to the routing layer **311**C. Each of the bond pads **322** may comprise a land grid array (LGA) pad, or the like, configured to be mounted on a print circuit board (PCB). To form the bond pads **322** and the bond pads **323**, a photoresist (not shown) may be formed over the routing structure **312** and patterned. Next a conductive material may be formed through a deposition process such as electroplating, electroless plating, or the like. The conductive material may comprise a metal (e.g., nickel, gold, the like, or a combination thereof). Once the conductive material has been formed, the photoresist may be removed through a suitable removal process such as ashing or chemical stripping. In some embodiments, the bond pads **322** may have a pitch P**1** that is in a range from about 200 μm to about 1000 μm. The bond pads **323** may have a pitch that is different from (e.g., smaller) the bond pads **322** to allow conductive connectors having different pitches to be formed on the bond pads **323**. For example, conductive connectors **506** formed on the bond pads **323** (described in detail below) may have a have a pitch P**2** that is in a range from about 50 μm to about 250 μm.
(31) A protection layer **320** is then formed over the routing structures **312** and **316** of interconnect structure **300**. The protection layer **320** may comprise a solder resist material or a PBO material, and may be formed to protect the surfaces of the routing structures **312** or **316**. In some embodiments, the protection layer **320** may be a photosensitive material formed by printing, lamination, spin-coating, or the like. The photosensitive material may then be exposed to an optical pattern and developed, forming openings in the photosensitive material that expose portions of the routing layer **315**C, the bond pads **322**, and the bond pads **323**. In other embodiments, the protection layer **320** may be formed by depositing a non-photosensitive dielectric layer (e.g., silicon oxide, silicon nitride, the like, or a combination), forming a patterned photoresist mask over the dielectric layer using suitable photolithography techniques, and then etching the dielectric layer using the patterned photoresist mask using a suitable etching process (e.g., wet etching or dry etching). The protection layer **320** may be formed and patterned over the routing structure **312** and the routing structure **316** using the same techniques. Other processes and materials may also be used. In some embodiments, the interconnect structure **300** may be formed with one or both of the routing structures **312** and **316**. In some embodiments, the interconnect structure **300** may be formed with the protection layer **320** formed and patterned over one or both of the routing structures **312** and **316**.
(32) Conductive connectors **506** are then formed on the exposed bond pads **323**. The conductive

connectors **506** are electrically coupled to the routing structure **312** through the bond pads **323**. The conductive connectors **506** may comprise ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **506** may comprise a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **506** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **506** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may comprise nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process. In some embodiments, the conductive connectors **506** may have the pitch P**2** that is in a range from about 50 μm to about 250 μm. In some embodiments, the bond pads **323** may have a pitch that is different from (e.g., smaller) the bond pads **322**.
(33) In FIG. **4**, the interconnect structure **300** is bonded to the redistribution structure **200**. In some embodiments, under-bump metallization structures (UBMs, not shown) are first formed on portions of the topmost redistribution layer of the redistribution structure **200** (e.g., redistribution layer **209**F in FIG. **2**). The UBMs may comprise three layers of conductive materials, such as a layer of titanium, a layer of copper, and a layer of nickel. However, other arrangements of materials and layers may be used, such as an arrangement of chrome/chrome-copper alloy/copper/gold, an arrangement of titanium/titanium tungsten/copper, or an arrangement of copper/nickel/gold, that are suitable for the formation of the UBMs. Any suitable materials or layers of material that may be used for the UBMs are fully intended to be included within the scope of the current application. The UBMs may be created by forming each layer of the UBMs over the redistribution structure **200**. The forming of each layer may be performed using a plating process (e.g., electroplating or electroless plating), sputtering, evaporation, PECVD, or the like. Once the layers of the UBMs have been formed, portions of the layers may then be removed through a suitable photolithographic masking and etching process.
(34) Still referring to FIG. **4**, conductive connectors **212** are then formed over the redistribution structure **200**. The conductive connectors **212** may be formed over the UBMs, if present. The conductive connectors **212** may comprise ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, contact bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **212** may comprise a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **212** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In some embodiments, the conductive connectors **212** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, the conductive connectors **212** may have a pitch P**3** that is in a range from about 100 μm to about 1000 μm.
(35) Next, a mounting process is performed to place the interconnect structure **300** into electrical connection with the redistribution structure **200**, in accordance with some embodiments. In an embodiment, the interconnect structure **300** is placed into physical contact with the conductive

connectors **212** on the redistribution structure **200** using, e.g., a pick and place process. The interconnect structure **300** may be placed such that exposed regions of a topmost routing layer (e.g., routing layers **315**C) are aligned with corresponding connectors of the conductive connectors **212** on the redistribution structure **200**. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **212** of the redistribution structure **200** to the interconnect structure **300**. In some embodiments, conductive connectors are formed on the interconnect structure **300** instead of or in addition to the conductive connectors **212** formed on the redistribution structure **200**. After the interconnect structure **300** is bonded to the redistribution structure **200**, a first gap **203** is formed between the protection layer **320** of the interconnect structure **300** and the top insulating layer (e.g., insulating layer **208**F) of the redistribution structure **200**. According to some embodiments, a height T**2** of the first gap **203** may be in a range from about 20 μm to about 500 μm.

(36) In FIG. **5**, a molding process is performed to encapsulate the interconnect structure **300** attached to the redistribution structure **200**. The molding process comprises depositing an underfill **402** to fill the first gap **203** in between the protection layer **320** of the interconnect structure **300** and the top insulating layer (e.g., insulating layer **208**F) of the redistribution structure **200**. According to some embodiments, the underfill **402** fills the voids within the first gap **203** in between conductive connectors **212** connecting the redistribution structure **200** and the interconnect structure **300** and forms along sidewalls of the interconnect structure **300** to align with an outer perimeter of the redistribution structure **200**. According to some embodiments, a thickness T**3** from a top surface of the protection layer **320** over the routing structures **312** to a bottom surface of the polymer layer **105** may be in a range from about 150 μm to about 5000 μm.

(37) According to some embodiments, the underfill **402** may comprise a molding compound, an epoxy, an underfill, a dispense molding underfill (DMUF), a resin, or the like. The underfill **402** may be dispensed using, e.g., a molding process, such as a transfer molding process, an injection process, combinations of these, or the like. The underfill **402** may protect the conductive connectors **212** and provide structural support for the redistribution structure **200**. In some embodiments, after dispensing the underfill **402** a curing process may be performed.

(38) In FIG. **6**A, a top-down view of the carrier substrate **102** is shown after the mounting of a plurality of interconnect structures **300** on the redistribution structure **200** using a wafer form process, in accordance with some embodiments. The carrier substrate **102** may have a round or circular shape and may be referred to herein as a wafer carrier substrate. The redistribution structure **200** may be larger and include a plurality of first package regions **1000**. For example, FIG. **6**A illustrates the redistribution structure **200** having a circular wafer shape with four first package regions **1000**A, **1000**B, **1000**C, and **1000**D included on the carrier substrate **102** allowing for four final package components to be fabricated on a single wafer and later singulated by sawing along lines **401** and around the outer edges of first package regions **1000**A, **1000**B, **1000**C, and **1000**D. As illustrated in FIG. **6**A, the first package regions **1000** may have rectangular shapes and may be formed on the carrier substrate **102** having a round shape, such as a circular shape. Although four of the first package regions **1000** are illustrated in FIG. **6**A, any number of the first package regions **1000** may be formed on the carrier substrate **102**.

(39) In FIG. **6**B, a top-down view of the carrier substrate **102** is shown after the mounting of a plurality of interconnect structures **300** on the redistribution structure **200** using a panel form process, in accordance with an alternate embodiment. The carrier substrate **102** may have a rectangular shape, and the carrier substrate **102** may be referred to as a panel. The redistribution structure **200** may be larger and include a plurality of first package regions **1000**. For example, FIG. **6**B illustrates the redistribution structure **200** having a rectangular shape with nine first package regions **1000** included on the carrier substrate **102** allowing for nine final package components to be fabricated on a single panel and later singulated by sawing along lines **401** and around the outer edges of the first package regions **1000**. Although nine of the first package regions

1000 are illustrated in FIG. **6**B, any number of the first package regions **1000** may be formed on the carrier substrate **102**.

(40) In FIG. **7**A, a packaged semiconductor device **400** is placed on the conductive connectors **506** of the interconnect structure **300**, making electrical connection between the packaged semiconductor device **400** and the routing structure **312** of the interconnect structure **300**. The packaged semiconductor device **400** may be placed on the conductive connectors **506** using a placement process such as a pick-and-place process, or the like. The packaged semiconductor device **400** may include one or more devices, which may include devices designed for an intended purpose such as a memory die (e.g., a DRAM die, a stacked memory die, a high-bandwidth memory (HBM) die, etc.), a logic die, a central processing unit (CPU) die, an I/O die, a system-on-a-chip (SoC), a component on a wafer (CoW), an integrated fan-out structure (InFO), a package, the like, or a combination thereof. In an embodiment, the packaged semiconductor device **400** includes integrated circuit devices, such as transistors, capacitors, inductors, resistors, metallization layers, conductive connectors, and the like, therein, as desired for a particular functionality. In some embodiments, the packaged semiconductor device **400** may include more than one of the same type of device, or may include different devices. FIG. **7**A shows three semiconductor devices encapsulated and connected with redistribution structures and contact pads, but in other embodiments one, two, or more than three semiconductor devices may be attached to the conductive connectors **506**.

(41) The packaged semiconductor device **400** shown in FIG. **7**A may comprise a plurality of the integrated circuit devices **510** and **512** that are attached to the interposer **504**. The interposer **504** may include a substrate **804** and the interconnect structure **802**. The integrated circuit devices **510** and **512** may each have a single function (e.g., a logic device, memory die, etc.), or may have multiple functions (e.g., a SoC). In an embodiment, the integrated circuit device **510** may comprise a SoC device, and the integrated circuit devices **512** may comprise I/O dies. The packaged semiconductor device **400** may comprise an underfill material **505** dispensed between the integrated circuit devices **510** and **512** and the interconnect structure **802**. In some embodiments, the packaged semiconductor device **400** may comprise an encapsulant **112** that surrounds the integrated circuit devices **510** and **512**.

(42) FIG. **7**B illustrates a cross-sectional view of the integrated circuit device **510**/**512** in accordance with some embodiments. The integrated circuit device **510**/**512** may be formed in a wafer, which may include different device regions that are singulated in subsequent steps to form a plurality of integrated circuit dies. The integrated circuit device **510**/**512** may be processed according to applicable manufacturing processes to form integrated circuits. For example, the integrated circuit device **510**/**512** includes a semiconductor substrate **52**, such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **52** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The semiconductor substrate **52** has an active surface (e.g., the surface facing upwards in FIG. **7**B), sometimes called a front side, and an inactive surface (e.g., the surface facing downwards in FIG. **7**B), sometimes called a back side.

(43) Devices (represented by a transistor) **54** may be formed at the front surface of the semiconductor substrate **52**. The devices **54** may be active devices (e.g., transistors, diodes, etc.), capacitors, resistors, etc. An inter-layer dielectric (ILD) **56** is over the front surface of the semiconductor substrate **52**. The ILD **56** surrounds and may cover the devices **54**. The ILD **56** may include one or more dielectric layers formed of materials such as Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), undoped Silicate Glass (USG), or the like.

(44) Conductive plugs **58** extend through the ILD **56** to electrically and physically couple the devices **54**. For example, when the devices **54** are transistors, the conductive plugs **58** may couple the gates and source/drain regions of the transistors. The conductive plugs **58** may be formed of tungsten, cobalt, nickel, copper, silver, gold, aluminum, the like, or combinations thereof. An interconnect structure **60** is over the ILD **56** and conductive plugs **58**. The interconnect structure **60** interconnects the devices **54** to form an integrated circuit. The interconnect structure **60** may be formed by, for example, metallization patterns in dielectric layers on the ILD **56**. The metallization patterns include metal lines and vias formed in one or more low-k dielectric layers. The metallization patterns of the interconnect structure **60** are electrically coupled to the devices **54** by the conductive plugs **58**.

(45) The integrated circuit device **510**/**512** further includes pads **62**, such as aluminum pads, to which external connections are made. The pads **62** are on the active side of the integrated circuit device **510**/**512**, such as in and/or on the interconnect structure **60**. One or more passivation films **64** are on the integrated circuit device **510**/**512**, such as on portions of the interconnect structure **60** and pads **62**. Openings extend through the passivation films **64** to the pads **62**. Die connectors **66**, such as conductive pillars (for example, formed of a metal such as copper), extend through the openings in the passivation films **64** and are physically and electrically coupled to respective ones of the pads **62**. The die connectors **66** may be formed by, for example, plating, or the like. The die connectors **66** electrically couple the respective integrated circuits of the integrated circuit device **510**/**512**.

(46) The packaged semiconductor device **400** may be placed such that conductive bumps **502** are aligned with the conductive connectors **506** of the interconnect structure **300**. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **506** of the interconnect structure **300** to the packaged semiconductor device **400**. In some embodiments, conductive connectors are formed on the packaged semiconductor device **400** instead of or in addition to the conductive connectors **506** formed on the interconnect structure **300**. An underfill **503** may then be formed between the packaged semiconductor device **400** and the interconnect structure **300**, surrounding the conductive connectors **506**. In some embodiments, the conductive connectors **506** are not formed on the interconnect structure **300**, and the packaged semiconductor device **400** is bonded to the interconnect structure **300** using a direct bonding technique such as thermocompression bonding, hybrid bonding, metal-to-metal bonding, or the like. However, any suitable bonding technique may be utilized.

(47) In FIG. **8**, a debonding process of the carrier substrate **102** and formation of conductive connectors **404** on the contact pads **104**, and the formation of conductive connectors **406** on the contact pads **106** of the redistribution structure **200** is shown. The carrier substrate **102** may be debonded from the redistribution structure **200** using, e.g., a thermal process to alter the adhesive properties of the adhesive layer **103** disposed on the carrier substrate **102**. In a particular embodiment an energy source such as an ultraviolet (UV) laser, a carbon dioxide (CO2) laser, or an infrared (IR) laser, is utilized to irradiate and heat the adhesive layer **103** until the adhesive layer **103** loses at least some of its adhesive properties. Once performed, the carrier substrate **102** and the adhesive layer **103** may be physically separated and removed from the redistribution structure **200**. Once the carrier substrate **102** and the adhesive layer **103** have been removed, the resulting structure may be flipped over and attached to a temporary substrate (not shown), such as a tape, wafer, panel, frame, ring, or the like for further processing.

(48) FIG. **8** additionally illustrates a patterning of the polymer layer **105** in order to expose the contact pads **104** and the contact pads **106**. In an embodiment, the polymer layer **105** may be patterned using, e.g., a laser drilling method. In such a method a protective layer, such as a light-to-heat conversion (LTHC) layer or a hogomax layer (not separately illustrated in FIG. **8**) is first deposited over the polymer layer **105**. Once protected, a laser is directed towards those portions of the polymer layer **105** which are desired to be removed in order to expose the underlying contact

pads **104** and the contact pads **106**.

(49) In another embodiment, the polymer layer **105** may be patterned by initially applying a photoresist (not individually illustrated in FIG. **8**) to the polymer layer **105** and then exposing the photoresist to a patterned energy source (e.g., a patterned light source) so as to induce a chemical reaction, thereby inducing a physical change in those portions of the photoresist exposed to the patterned light source. A developer is then applied to the exposed photoresist to take advantage of the physical changes and selectively remove either the exposed portion of the photoresist or the unexposed portion of the photoresist, depending upon the desired pattern, and the underlying exposed portion of the polymer layer **105** are removed with, e.g., a dry etch process. However, any other suitable method for patterning the polymer layer **105** may be utilized.

(50) Once the contact pads **104** and the contact pads **106** have been exposed, the conductive connectors **404** may be formed over the contact pads **104**, and the conductive connectors **406** may be formed over the contact pads **106** making electrical connection to the redistribution structure **200**. In some embodiments, an optional solderability treatment (e.g., pre-soldering treatment) may be performed on the exposed surfaces of the contact pads **104** and the contact pads **106** prior to forming the conductive connectors **404** and **406**. The conductive connectors **404** and the conductive connectors **406** may comprise ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **404** and **406** may comprise a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **404** and **406** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In some embodiments, the conductive connectors **404** may be similar to conductive connectors **506** (described previously in FIG. **3**). In some embodiments, the conductive connectors **404** formed on the contact pads **104** and the conductive connectors **406** formed on the contact pads **106** may be different. For example, in an embodiment, the conductive connectors **404** formed on the contact pads **104** may comprise controlled collapse chip connection (C4) bumps and the conductive connectors **406** formed on the contact pads **106** may comprise ball grid array (BGA) connectors. In some embodiments, the conductive connectors **404** may have a pitch P**4** that is in a range from about 50 μm to about 250 μm. In some embodiments, the conductive connectors **406** may have a pitch P**5** that is in a range from about 150 μm to about 1000 μm. In some embodiments, the pitch P**4** of the conductive connectors **404** may be the same or different from the pitch P**2** of the conductive connectors **506** formed on the bond pads **323** (described above in FIG. **3**).

(51) After the formation of the conductive connectors **404** and the conductive connectors **406**, a singulation process may be performed by sawing along scribe line regions, for example, between the first package regions **1000** by sawing along lines **401** and around the outer edges of first package regions **1000**A, **1000**B, **1000**C, and **1000**D as shown above in FIG. **6**A, or between the first package regions **1000** by sawing along lines **401** and around the outer edges of the first package regions **1000** as shown above in FIG. **6**B. The sawing singulates each first package region **1000** from adjacent first package regions **1000** to form multiple singulated package components.

(52) FIG. **9** illustrates a cross-sectional view of intermediate steps in the forming of the device package **2500**. In FIG. **9**, a packaged semiconductor device **500** is placed on the conductive connectors **404** of the redistribution structure **200**, making electrical connection between the packaged semiconductor device **500** and the contact pads **104** of the redistribution structure **200**. Unless specified otherwise, like reference numerals in the packaged semiconductor device **500** represent like components in the packaged semiconductor device **400** (described previously in FIGS. **7**A and **7**B). Accordingly, the process steps and applicable materials may not be repeated

herein. The packaged semiconductor device **500** may be placed on the conductive connectors **404** using a placement process such as a pick-and-place process, or the like. The packaged semiconductor device **500** may include one or more devices, which may include devices designed for an intended purpose such as a memory die (e.g., a DRAM die, a stacked memory die, a high-bandwidth memory (HBM) die, etc.), a logic die, a central processing unit (CPU) die, an I/O die, a system-on-a-chip (SoC), a component on a wafer (CoW), an integrated fan-out structure (InFO), a package, the like, or a combination thereof. In an embodiment, the packaged semiconductor device **500** includes integrated circuit devices, such as transistors, capacitors, inductors, resistors, metallization layers, conductive connectors, and the like, therein, as desired for a particular functionality. In some embodiments, the packaged semiconductor device **500** may include more than one of the same type of device, or may include different devices. FIG. **9** shows three semiconductor devices encapsulated and connected with redistribution structures and contact pads, but in other embodiments one, two, or more than three semiconductor devices may be attached to the conductive connectors **404**.

(53) The packaged semiconductor device **500** shown in FIG. **9** may comprise a plurality of the integrated circuit devices **520** and **522**. The integrated circuit devices **520** and **522** may each have a single function (e.g., a logic device, memory die, etc.), or may have multiple functions (e.g., a SoC). In an embodiment, the integrated circuit device **520** may be similar or different to the integrated circuit device **510**. In an embodiment the integrated circuit devices **522** maybe similar or different to the integrated circuit devices **512**. In an embodiment, the integrated circuit device **520** may comprise a CPU device, and the integrated circuit devices **512** may comprise HBM dies.

(54) Still referring to FIG. **9**, one or more packaged semiconductor devices **600** is placed on the conductive connectors **406** of the redistribution structure **200**, making electrical connection between the packaged semiconductor devices **600** and the contact pads **106** of the redistribution structure **200**. The packaged semiconductor devices **600** may also be referred to as integrated fan-out (InFO) packages. Each of the packaged semiconductor devices **600** may comprise one or more integrated circuit dies **528** that are electrically connected to the redistribution structure **200** through the front-side redistribution structure **530**. The front-side redistribution structure **530** includes any number of dielectric layers and any number of metallization patterns. The metallization patterns may also be referred to as redistribution layers or redistribution lines. Each integrated circuit die **528** may be a logic die (e.g., central processing unit (CPU), graphics processing unit (GPU), system-on-a-chip (SoC), application processor (AP), microcontroller, etc.), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management die (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies), the like, or combinations thereof. The packaged semiconductor devices **600** may be placed on the conductive connectors **406** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **406** of the redistribution structure **200** to the packaged semiconductor devices **600**.

(55) Advantages can be achieved as a result of the formation of the device package **2500** in which component devices (e.g., packaged semiconductor device **400**, packaged semiconductor device **500**, packaged semiconductor device **600**, or the like) are bonded on opposite sides of the core substrate **302** to form a system on integrated substrate (SoIS). The core substrate **302** includes an organic substrate, and the core substrate **302** may be part of the pre-fabricated interconnect structure **300** that comprises pre-soldered conductive connectors **404**. These advantages may include the ability to electrically connect and integrate a higher number of component devices (e.g., packaged semiconductor device **400**, packaged semiconductor device **500**, packaged semiconductor device **600** . . .) within the device package **2500**. In addition, because of the compact structure of the device package **2500**, the package components (e.g., packaged

semiconductor device **400**, packaged semiconductor device **500**, packaged semiconductor device **600**, or the like) are closer to each other which results in better power and signal integrity with reduced transmission losses. Accordingly, this leads to better computing performance and allows the use of the device package in applications that require faster processing of data and performing of complex calculations at high speeds (e.g., High-Performance-Computing (HPC)).

(56) FIG. **10** illustrates a cross-sectional view of the device package **2500** in accordance with some embodiments. In FIG. **10**, the structure shown in FIG. **9** is mounted on to a support substrate **135** (e.g., a printed circuit board (PCB)). The support substrate **135** may be a printed circuit board such as a laminate substrate formed as a stack of multiple thin layers (or laminates) of a polymer material such as bismaleimide triazine (BT), FR-4, ABF, or the like. However, any other suitable substrate, such as a silicon interposer, a silicon substrate, organic substrate, a ceramic substrate, or the like, may also be utilized. Conductive connectors **136** may be used to bond the support substrate **135** to the bond pads **322** of the interconnect structure **300** as illustrated in FIG. **10**. The conductive connectors **136** may comprise metal pins that can be connected to socket connectors on the support substrate **135**. The metal pins may be formed using an electroless nickel-electroless palladium-immersion gold technique (ENEPIG), or the like. The conductive connectors **136** may have a height H**1** that is larger than a height H**2** of the packaged semiconductor device **400** to allow the packaged semiconductor device **400** to be disposed between the support substrate **135** and the interconnect structure **300**. In other embodiments, the conductive connectors **136** may be electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like.

(57) In other embodiments, the conductive connectors **136** may include a conductive material such as solder, copper, copper pin, alloy pin, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. The conductive connectors **136** may be first formed on either the support substrate **135**, or the interconnect structure **300**, and then reflowed to complete the bond. In such an embodiment, the height H**1** of the conductive connectors **136** may be less than the height H**2** of the packaged semiconductor device **400**, and the support substrate **135** may comprise a cavity to accommodate the packaged semiconductor device **400**.

(58) FIG. **11** illustrates a cross-sectional view of intermediate steps in the forming of the device package **5000**. The device package **5000** is an alternative embodiment in which like reference numerals represent like components in the embodiment shown in FIGS. **1** through **9**, unless specified otherwise. Accordingly, the process steps and applicable materials may not be repeated herein. The initial steps of this embodiment are essentially the same as shown in FIGS. **1** through **5**. As shown in FIG. **11**, a packaged semiconductor device **600** is placed on the conductive connectors **506** of the interconnect structure **300**, making electrical connection between the packaged semiconductor device **600** and the routing structure **312** of the interconnect structure **300**. The packaged semiconductor device **600** may be placed on the conductive connectors **506** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **506** of the interconnect structure **300** to the packaged semiconductor device **600** through the front-side redistribution structure **530**. In some embodiments, conductive connectors are formed on the packaged semiconductor device **600** instead of or in addition to the conductive connectors **506** formed on the interconnect structure **300**. An underfill **503** may then be formed between the packaged semiconductor device **600** and the interconnect structure **300**, surrounding the conductive connectors **506**. In some embodiments, the conductive connectors **506** are not formed on the interconnect structure **300**, and the packaged semiconductor device **600** is bonded to the interconnect structure **300** using a direct bonding technique such as thermocompression bonding, hybrid bonding, metal-to-metal bonding, or the like. However, any suitable bonding technique may be utilized.

(59) Next, a debonding process of the carrier substrate **102** and formation of conductive connectors **404** on the contact pads **104**, and the formation of conductive connectors **406** on the contact pads

106 of the redistribution structure **200** are performed. The process steps and applicable materials are essentially the same as those shown and described previously in FIG. **8**. Accordingly, the process steps and applicable materials may not be repeated herein.
(60) Still referring to FIG. **11**, a packaged semiconductor device **600** is placed on the conductive connectors **404** of the redistribution structure **200**, making electrical connection between the packaged semiconductor device **600** and the contact pads **104** of the redistribution structure **200** through the front-side redistribution structure **530**. The packaged semiconductor device **600** may be placed on the conductive connectors **404** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **404** of the redistribution structure **200** to the packaged semiconductor device **600**. In some embodiments, conductive connectors are formed on the packaged semiconductor device **600** instead of or in addition to the conductive connectors **404** formed on the redistribution structure **200**. The underfill **503** may then be formed between the packaged semiconductor device **600** and the redistribution structure **200**, surrounding the conductive connectors **404**. In some embodiments, the conductive connectors **404** are not formed on the redistribution structure **200**, and the packaged semiconductor device **600** is bonded to the redistribution structure **200** using a direct bonding technique such as thermocompression bonding, hybrid bonding, metal-to-metal bonding, or the like. However, any suitable bonding technique may be utilized.
(61) Next, one or more packaged semiconductor devices **600** are placed on the conductive connectors **406** of the redistribution structure **200**, making electrical connection between the packaged semiconductor devices **600** and the contact pads **106** of the redistribution structure **200**. The packaged semiconductor devices **600** may be placed on the conductive connectors **406** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **406** of the redistribution structure **200** to the packaged semiconductor devices **600**.
(62) FIG. **12** illustrates a cross-sectional view of the device package **5000** in accordance with some embodiments. In FIG. **12**, the structure shown in FIG. **11** is mounted on to a support substrate **135** (e.g., a printed circuit board (PCB)). The support substrate **135** may be a printed circuit board such as a laminate substrate formed as a stack of multiple thin layers (or laminates) of a polymer material such as bismaleimide triazine (BT), FR-4, ABF, or the like. However, any other suitable substrate, such as a silicon interposer, a silicon substrate, organic substrate, a ceramic substrate, or the like, may also be utilized. Conductive connectors **136** may be used to bond the support substrate **135** to the bond pads **322** of the interconnect structure **300** as illustrated in FIG. **12**. The conductive connectors **136** may comprise metal pins that can be connected to socket connectors on the support substrate **135**. The metal pins may be formed using an electroless nickel-electroless palladium-immersion gold technique (ENEPIG), or the like. The conductive connectors **136** may have a height H**3** that is larger than a height H**4** of the packaged semiconductor device **600** to allow the packaged semiconductor device **600** to be disposed between the support substrate **135** and the interconnect structure **300**. In other embodiments, the conductive connectors **136** may be electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. In other embodiments, the conductive connectors **136** may include a conductive material such as solder, copper, copper pin, alloy pin, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. The conductive connectors **136** may be first formed on either the support substrate **135**, or the interconnect structure **300**, and then reflowed to complete the bond. In such an embodiment, the height H**3** of the conductive connectors **136** may be less than the height H**4** of the packaged semiconductor device **600**, and the support substrate **135** may comprise a cavity to accommodate the packaged semiconductor device **600**.
(63) FIG. **13** illustrates a cross-sectional view of intermediate steps in the forming of the device package **7500**. The device package **7500** is an alternative embodiment in which like reference numerals represent like components in the embodiment shown in FIGS. **1** through **9**, unless

specified otherwise. Accordingly, the process steps and applicable materials may not be repeated herein. The initial steps of this embodiment are essentially the same as shown in FIGS. **1** through **2**. Next, a mounting process is performed to place an interconnect structure **700** into electrical connection with the redistribution structure **200**, and a molding process is performed to encapsulate the interconnect structure **700** attached to the redistribution structure **200**, in accordance with some embodiments. The mounting process and the molding process are similar to those described earlier in FIGS. **4** and **5**, unless specified otherwise. Accordingly, the process steps and applicable materials may not be repeated herein.

(64) The interconnect structure **700** is formed in a separate process, and can be tested separately so that a known good interconnect structure **700** is used. The interconnect structure **700** is subsequently bonded to the redistribution structure **200**. Like reference numerals in the interconnect structure **700** represent like components in the interconnect structure **300** (described previously in FIG. **3**), unless specified otherwise. Accordingly, the process steps and applicable materials may not be repeated herein. During the formation of the interconnect structure **700**, dielectric layers **314**A-C and routing layers **315**A-C may be formed adjacent to the routing layer **309** to form the routing structure **316**, and dielectric layers **310**A-C and routing layers **311**A-C may be formed adjacent to the routing layer **308** to form the routing structure **312**, according to some embodiments. In some embodiments, bond pads **322** and contact pads **107** are formed over and electrically connected to the routing layer **311**C. Each of the bond pads **322** may comprise a land grid array (LGA) pad, or the like, configured to be mounted on a print circuit board (PCB). To form the bond pads **322**, a photoresist (not shown) may be formed over the routing structure **312** and patterned. Next a conductive material may be formed through a deposition process such as electroplating, electroless plating, or the like. The conductive material may comprise a metal (e.g., nickel, gold, the like, or a combination thereof). Once the conductive material has been formed, the photoresist may be removed through a suitable removal process such as ashing or chemical stripping. In some embodiments, the bond pads **322** may have a pitch P**6** that is in a range from about 200 μm to about 1000 μm. To form the contact pads **107**, a photoresist (also not shown) may be formed and patterned to cover the routing structure **312**, e.g., a spin coating technique. Once the photoresist has been formed and patterned, a conductive material may be formed that may be a material such as copper, titanium, tungsten, aluminum, another metal, the like, or a combination thereof. The conductive material may be formed through a deposition process such as electroplating, electroless plating, or the like. Once the conductive material has been formed, the photoresist may be removed through a suitable removal process such as ashing or chemical stripping.

(65) Further, during the formation of the interconnect structure **700**, a protection layer **320** is then formed over the routing structures **312** and **316** of interconnect structure **700**. The protection layer **320** may comprise a solder resist material or a PBO material, and may be formed to protect the surfaces of the routing structures **312** or **316**. In some embodiments, the protection layer **320** may be a photosensitive material formed by printing, lamination, spin-coating, or the like. The photosensitive material may then be exposed to an optical pattern and developed, forming openings in the photosensitive material that expose portions of the bond pads **322**, the routing layer **315**C, and the contact pads **107**. In other embodiments, the protection layer **320** may be formed by depositing a non-photosensitive dielectric layer (e.g., silicon oxide, silicon nitride, the like, or a combination), forming a patterned photoresist mask over the dielectric layer using suitable photolithography techniques, and then etching the dielectric layer using the patterned photoresist mask using a suitable etching process (e.g., wet etching or dry etching). The protection layer **320** may be formed and patterned over the routing structure **312** and the routing structure **316** using the same techniques. Other processes and materials may also be used. In some embodiments, the interconnect structure **700** may be formed with one or both of the routing structures **312** and **316**. In some embodiments, the interconnect structure **700** may be formed with the protection layer **320**

formed and patterned over one or both of the routing structures **312** and **316**.

(66) After the mounting process is performed to place the interconnect structure **700** into electrical connection with the redistribution structure **200**, and the molding process is performed to encapsulate the interconnect structure **700** attached to the redistribution structure **200**, the conductive connectors **408** may be formed over the contact pads **107**, making electrical connection to the interconnect structure **700**. In some embodiments the conductive connectors **408** may be formed over the contact pads **107** before the interconnect structure **700** is bonded to the redistribution structure **200**. In some embodiments, an optional solderability treatment (e.g., pre-soldering treatment) may be performed on the exposed surfaces of the contact pads **107** prior to forming the conductive connectors **408**. The conductive connectors **408** may comprise ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **408** may comprise a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **408** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In some embodiments, the conductive connectors **408** may have a pitch P**7** that is in a range from about 150 μm to about 1000 μm.

(67) Next, one or more packaged semiconductor devices **600** are placed on the conductive connectors **408** of the interconnect structure **700**, making electrical connection between the packaged semiconductor devices **600** and the contact pads **107** of the interconnect structure **700**. Unless specified otherwise, like reference numerals in the packaged semiconductor device **600** represent like components in the packaged semiconductor device **600** described previously in FIG. **9**. Accordingly, the process steps and applicable materials may not be repeated herein. The packaged semiconductor devices **600** may be placed on the conductive connectors **408** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **408** of the interconnect structure **700** to the packaged semiconductor devices **600**.

(68) In FIG. **14**, the carrier substrate **102** may be debonded from the redistribution structure **200** using, e.g., a thermal process to alter the adhesive properties of the adhesive layer **103** disposed on the carrier substrate **102**. In a particular embodiment an energy source such as an ultraviolet (UV) laser, a carbon dioxide (CO2) laser, or an infrared (IR) laser, is utilized to irradiate and heat the adhesive layer **103** until the adhesive layer **103** loses at least some of its adhesive properties. Once performed, the carrier substrate **102** and the adhesive layer **103** may be physically separated and removed from the redistribution structure **200**. Once the carrier substrate **102** and the adhesive layer **103** have been removed, the resulting structure may be flipped over and attached to a temporary substrate (not shown), such as a tape, wafer, panel, frame, ring, or the like for further processing.

(69) FIG. **14** additionally illustrates a patterning of the polymer layer **105** in order to expose the contact pads **104** and the contact pads **106**. In an embodiment, the polymer layer **105** may be patterned using, e.g., a laser drilling method. In such a method a protective layer, such as a light-to-heat conversion (LTHC) layer or a hogomax layer (not separately illustrated in FIG. **14**) is first deposited over the polymer layer **105**. Once protected, a laser is directed towards those portions of the polymer layer **105** which are desired to be removed in order to expose the underlying contact pads **104** and the contact pads **106**.

(70) In another embodiment, the polymer layer **105** may be patterned by initially applying a photoresist (not individually illustrated in FIG. **14**) to the polymer layer **105** and then exposing the photoresist to a patterned energy source (e.g., a patterned light source) so as to induce a chemical reaction, thereby inducing a physical change in those portions of the photoresist exposed to the

patterned light source. A developer is then applied to the exposed photoresist to take advantage of the physical changes and selectively remove either the exposed portion of the photoresist or the unexposed portion of the photoresist, depending upon the desired pattern, and the underlying exposed portion of the polymer layer **105** are removed with, e.g., a dry etch process. However, any other suitable method for patterning the polymer layer **105** may be utilized.

(71) Once the contact pads **104** and the contact pads **106** have been exposed, the conductive connectors **404** may be formed over the contact pads **104**, and the conductive connectors **406** may be formed over the contact pads **106** making electrical connection to the redistribution structure **200**. In some embodiments, an optional solderability treatment (e.g., pre-soldering treatment) may be performed on the exposed surfaces of the contact pads **104** and the contact pads **106** prior to forming the conductive connectors **404** and **406**. The conductive connectors **404** and the conductive connectors **406** may comprise ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **404** and **406** may comprise a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **404** and **406** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In some embodiments, the conductive connectors **406** may be similar to conductive connectors **408** (described previously in FIG. **13**). In some embodiments, the conductive connectors **404** formed on the contact pads **104** and the conductive connectors **406** formed on the contact pads **106** may be different. For example, in an embodiment, the conductive connectors **404** formed on the contact pads **104** may comprise controlled collapse chip connection (C4) bumps and the conductive connectors **406** formed on the contact pads **106** may comprise ball grid array (BGA) connectors. In some embodiments, the conductive connectors **406** may have a pitch P**8** that is in a range from about 150 μm to about 1000 μm.

(72) Next, a packaged semiconductor device **800** is placed on the conductive connectors **404** of the redistribution structure **200**, making electrical connection between the packaged semiconductor device **800** and the contact pads **104** of the redistribution structure **200**. Unless specified otherwise, like reference numerals in the packaged semiconductor device **800** represent like components in the packaged semiconductor device **400** (described previously in FIGS. **7**A and **7**B). Accordingly, the process steps and applicable materials may not be repeated herein. The packaged semiconductor device **800** may be placed on the conductive connectors **404** using a placement process such as a pick-and-place process, or the like. The packaged semiconductor device **800** may include one or more devices, which may include devices designed for an intended purpose such as a memory die (e.g., a DRAM die, a stacked memory die, a high-bandwidth memory (HBM) die, etc.), a logic die, a central processing unit (CPU) die, an I/O die, a system-on-a-chip (SoC), a component on a wafer (CoW), an integrated fan-out structure (InFO), a package, the like, or a combination thereof. In an embodiment, the packaged semiconductor device **800** includes integrated circuit devices, such as transistors, capacitors, inductors, resistors, metallization layers, conductive connectors, and the like, therein, as desired for a particular functionality. In some embodiments, the packaged semiconductor device **800** may include more than one of the same type of device, or may include different devices. FIG. **14** shows three semiconductor devices encapsulated and connected with redistribution structures and contact pads, but in other embodiments one, two, or more than three semiconductor devices may be attached to the conductive connectors **404**.

(73) The packaged semiconductor device **800** shown in FIG. **14** may comprise a plurality of the integrated circuit devices **550** and **552**. The integrated circuit devices **550** and **552** may each have a single function (e.g., a logic device, memory die, etc.), or may have multiple functions (e.g., a SoC). In an embodiment, the integrated circuit device **550** may be similar or different to the

integrated circuit device **510** (described previously in FIG. **7**). In an embodiment the integrated circuit devices **552** maybe similar or different to the integrated circuit devices **512** (described previously in FIGS. **7**A and **7**B). In an embodiment, the integrated circuit device **550** may comprise a CPU device, and the integrated circuit devices **552** may comprise I/O dies.

(74) Still referring to FIG. **14**, one or more packaged semiconductor devices **600** are placed on the conductive connectors **406** of the redistribution structure **200**, making electrical connection between the packaged semiconductor devices **600** and the contact pads **106** of the redistribution structure **200**. Unless specified otherwise, like reference numerals in the packaged semiconductor device **600** represent like components in the packaged semiconductor device **600** described previously in FIG. **9**. Accordingly, the process steps and applicable materials may not be repeated herein. The packaged semiconductor devices **600** may be placed on the conductive connectors **406** using a placement process such as a pick-and-place process, or the like. Once in physical contact, a reflow process may be utilized to bond the conductive connectors **406** of the redistribution structure to the packaged semiconductor devices **600**.

(75) FIG. **15** illustrates a cross-sectional view of the device package **7500** in accordance with some embodiments. In FIG. **15**, the structure shown in FIG. **14** is mounted on to a support substrate **135** (e.g., a printed circuit board (PCB)). The support substrate **135** may be a printed circuit board such as a laminate substrate formed as a stack of multiple thin layers (or laminates) of a polymer material such as bismaleimide triazine (BT), FR-4, ABF, or the like. However, any other suitable substrate, such as a silicon interposer, a silicon substrate, organic substrate, a ceramic substrate, or the like, may also be utilized. Conductive connectors **136** may be used to bond the support substrate **135** to the bond pads **322** of the interconnect structure **700** as illustrated in FIG. **15**. The conductive connectors **136** may comprise metal pins that can be connected to socket connectors on the support substrate **135**. The metal pins may be formed using an electroless nickel-electroless palladium-immersion gold technique (ENEPIG), or the like. The conductive connectors **136** may have a height H**5** that is larger than a height H**6** of the one or more packaged semiconductor devices **600** to allow the one or more packaged semiconductor devices **600** to be disposed between the support substrate **135** and the interconnect structure **300**. In other embodiments, the conductive connectors **136** may be electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. In other embodiments, the conductive connectors **136** may include a conductive material such as solder, copper, copper pin, alloy pin, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. The conductive connectors **136** may be first formed on either the support substrate **135**, or the interconnect structure **300**, and then reflowed to complete the bond. In such an embodiment, the height H**5** of the conductive connectors **136** may be less than the height H**6** of the packaged semiconductor devices **600**, and the support substrate **135** may comprise one or more cavities to accommodate the packaged semiconductor devices **600**.

(76) The embodiments of the present disclosure have some advantageous features. The embodiments include the formation of a device package in which component devices (e.g., integrated fan-out (InFO) packages, chip-on-wafer (CoW) packages, or the like) are bonded on opposite sides of a core substrate to form a system on integrated substrate (SoIS). The core substrate may include an organic substrate, and the core substrate may be part of a pre-fabricated interconnect structure that comprises pre-soldered controlled collapse chip connection (C4) bumps. As a result, one or more embodiments disclosed herein may include the ability to electrically connect and integrate a higher number of component devices within the device package. In addition, because of the compact structure of the device package, the package components are closer to each other which results in better power and signal integrity with reduced transmission losses. This leads to better computing performance and allows the use of the device package in applications that require faster processing of data and performing of complex calculations at high speeds (e.g., High-Performance-Computing (HPC)).

(77) In accordance with an embodiment, a method includes forming a redistribution structure on a carrier substrate; coupling a first side of a first interconnect structure to a first side of the redistribution structure using first conductive connectors, where the first interconnect structure includes a core substrate, where the first interconnect structure includes second conductive connectors on a second side of the first interconnect structure opposite the first side of the first interconnect structure; coupling a first semiconductor device to the second side of the first interconnect structure using the second conductive connectors; removing the carrier substrate; and coupling a second semiconductor device to a second side of the redistribution structure using third conductive connectors, where the second side of the redistribution structure is opposite the first side of the redistribution structure. In an embodiment, the core substrate includes an organic material. In an embodiment, coupling the first side of the first interconnect structure to the first side of the redistribution structure includes performing a reflow process on the first conductive connectors. In an embodiment, the coupling the first side of the first interconnect structure to the first side of the redistribution structure results in a first gap between the first interconnect structure and the redistribution structure. In an embodiment, the method further includes depositing an underfill material to fill the first gap between the first interconnect structure and the redistribution structure. In an embodiment, the underfill material physically contacts sidewalls of the first interconnect structure. In an embodiment, the method further includes forming fourth conductive connectors on the second side of the redistribution structure; and coupling a third semiconductor device to the second side of the redistribution structure using the fourth conductive connectors. In an embodiment, the first conductive connectors and the fourth conductive connectors include ball grid array (BGA) connectors and the second conductive connectors and the third conductive connectors include controlled collapse chip connection (C4) bumps. In an embodiment, a second pitch of the second conductive connectors is smaller than a first pitch of the first conductive connectors and a third pitch of the fourth conductive connectors.
(78) In accordance with an embodiment, a method includes forming a redistribution structure on a carrier substrate, where a bottomost layer of the redistribution structure includes first contact pads and second contact pads, where the first contact pads have a different pitch than the second contact pads; mounting a first interconnect structure over the redistribution structure, where the first interconnect structure is electrically coupled to the redistribution structure through first conductive connectors on a first side of the first interconnect structure, where the first interconnect structure includes a core substrate, where the first interconnect structure includes second conductive connectors and bond pads on a second side of the first interconnect structure opposite the first side of the first interconnect structure; electrically coupling a first semiconductor device to the first interconnect structure through the second conductive connectors; and electrically coupling a second semiconductor device to the first contact pads of the redistribution structure through third conductive connectors, the first semiconductor device being located on an opposite side of the first interconnect structure than the second semiconductor device. In an embodiment, a first pitch of the bond pads and a second pitch of the second conductive connectors is different. In an embodiment, the method further includes coupling a printed circuit board to the bond pads of the first interconnect structure using fourth conductive connectors. In an embodiment, a first height of the fourth conductive connectors is larger than a second height of the first semiconductor device. In an embodiment, the method further includes electrically coupling a third semiconductor device to the first interconnect structure through the second conductive connectors, the first semiconductor device and the third semiconductor device being located on an opposite side of the first interconnect structure than the second semiconductor device. In an embodiment, the method further includes coupling a printed circuit board to the bond pads of the first interconnect structure using fifth conductive connectors, where the fifth conductive connectors are between the first semiconductor device and the third semiconductor device.
(79) In accordance with an embodiment, a semiconductor package includes a redistribution

structure; a first interconnect structure electrically connected to a first side of the redistribution structure, where the first interconnect structure includes a core substrate; an underfill material between the first interconnect structure and the redistribution structure; a first semiconductor package bonded to a second side of the redistribution structure; and a second semiconductor package bonded to the first interconnect structure, the second semiconductor package being located on an opposite side of the first interconnect structure than the first semiconductor package. In an embodiment, the underfill material further physically contacts sidewalls of the first interconnect structure. In an embodiment, further including a third semiconductor package bonded to the second side of the redistribution structure by first conductive connectors. In an embodiment, the first semiconductor package is bonded to the second side of the redistribution structure by second conductive connectors, and the second semiconductor package is bonded to the first interconnect structure by third conductive connectors, where a first pitch of the first conductive connectors is different from a second pitch of the second conductive connectors and the third conductive connectors. In an embodiment, further including a printed circuit board coupled to the first interconnect structure by fourth conductive connectors, where a first height of the fourth conductive connectors is larger than a second height of the second semiconductor package.
(80) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: coupling a first side of a first interconnect structure to a first side of a redistribution structure using first conductive connectors, wherein the first interconnect structure comprises a core substrate having conductive layers disposed on opposite surfaces, wherein the core substrate comprises an organic material; coupling a first semiconductor device to a second side of the first interconnect structure using second conductive connectors; and coupling a printed circuit board to the second side of the first interconnect structure using third conductive connectors, wherein a height of the third conductive connectors is larger than a height of the first semiconductor device.
2. The method of claim 1, further comprising coupling a second semiconductor device to a second side of the redistribution structure using fourth conductive connectors, wherein the second semiconductor device overlaps the first semiconductor device.
3. The method of claim 2, further comprising coupling a third semiconductor device to the second side of the redistribution structure using fifth conductive connectors, the third semiconductor device being adjacent to the second semiconductor device, and wherein the third semiconductor device overlaps at least one of the third conductive connectors.
4. The method of claim 3, wherein a first pitch of the second conductive connectors, and a second pitch of the fourth conductive connectors are smaller than a third pitch of the first conductive connectors and a fourth pitch of the fifth conductive connectors.
5. The method of claim 1, wherein the third conductive connectors comprise metal pins.
6. The method of claim 1, wherein after coupling the first side of the first interconnect structure to the first side of the redistribution structure, a gap is disposed between the first interconnect structure and the redistribution structure.
7. The method of claim 6, wherein the gap has a height that is in a range from 20 μm to 500 μm.

8. A semiconductor package comprising: a first interconnect structure electrically coupled to a first side of a redistribution structure, the first interconnect structure comprising a core substrate, wherein the core substrate comprises an organic material; a first semiconductor package bonded to a second side of the redistribution structure; a second semiconductor package bonded to the first interconnect structure, wherein the first interconnect structure is disposed between the redistribution structure and the second semiconductor package; and a support substrate coupled to the first interconnect structure by first conductive connectors and second conductive connectors, wherein the first conductive connectors and the second conductive connectors comprise metal pins, wherein a first conductive connector of the first conductive connectors is disposed adjacent to a first sidewall of the second semiconductor package, and a second conductive connector of the second conductive connectors is disposed adjacent to a second sidewall of the second semiconductor package.
9. The semiconductor package of claim 8, further comprising an underfill material disposed between the first interconnect structure and the redistribution structure.
10. The semiconductor package of claim 9, wherein the underfill material is in physical contact with sidewalls of the first interconnect structure.
11. The semiconductor package of claim 8, further comprising a third semiconductor package bonded to the second side of the redistribution structure.
12. The semiconductor package of claim 8, wherein top surfaces of the first conductive connectors are above a top surface of the second semiconductor package, and bottom surfaces of the first conductive connectors are below a bottom surface of the second semiconductor package.
13. A method comprising: forming a redistribution structure on a carrier substrate; coupling a first side of a first interconnect structure to a first side of the redistribution structure using first conductive connectors; coupling a first semiconductor device to a center portion of a second side of the first interconnect structure using second conductive connectors; coupling a second semiconductor device to a center portion of a second side of the redistribution structure using third conductive connectors; and coupling a printed circuit board to the second side of the first interconnect structure using fourth conductive connectors and fifth conductive connectors, wherein the first semiconductor device is disposed between the fourth conductive connectors and the fifth conductive connectors.
14. The method of claim 13, further comprising coupling a third semiconductor device to the second side of the redistribution structure using sixth conductive connectors, wherein the third semiconductor device overlaps the fourth conductive connectors.
15. The method of claim 14, further comprising coupling a fourth semiconductor device to the second side of the redistribution structure using seventh conductive connectors, wherein the fourth semiconductor device overlaps the fifth conductive connectors.
16. The method of claim 15, wherein the fourth conductive connectors and the fifth conductive connectors comprise metal pins.
17. The method of claim 13, wherein a height of the fourth conductive connectors and the fifth conductive connectors is larger than a height of the first semiconductor device.
18. The method of claim 13, further comprising: after coupling the first side of the first interconnect structure to the first side of the redistribution structure, depositing an underfill material to fill a gap between the first interconnect structure and the redistribution structure.
19. The method of claim 18, wherein after depositing the underfill material to fill the gap, the underfill material is in physical contact with sidewalls of the first interconnect structure.
20. The method of claim 19, wherein the gap has a height that is in a range from 20 μm to 500 μm.